



DS & AI

Artificial Intelligence

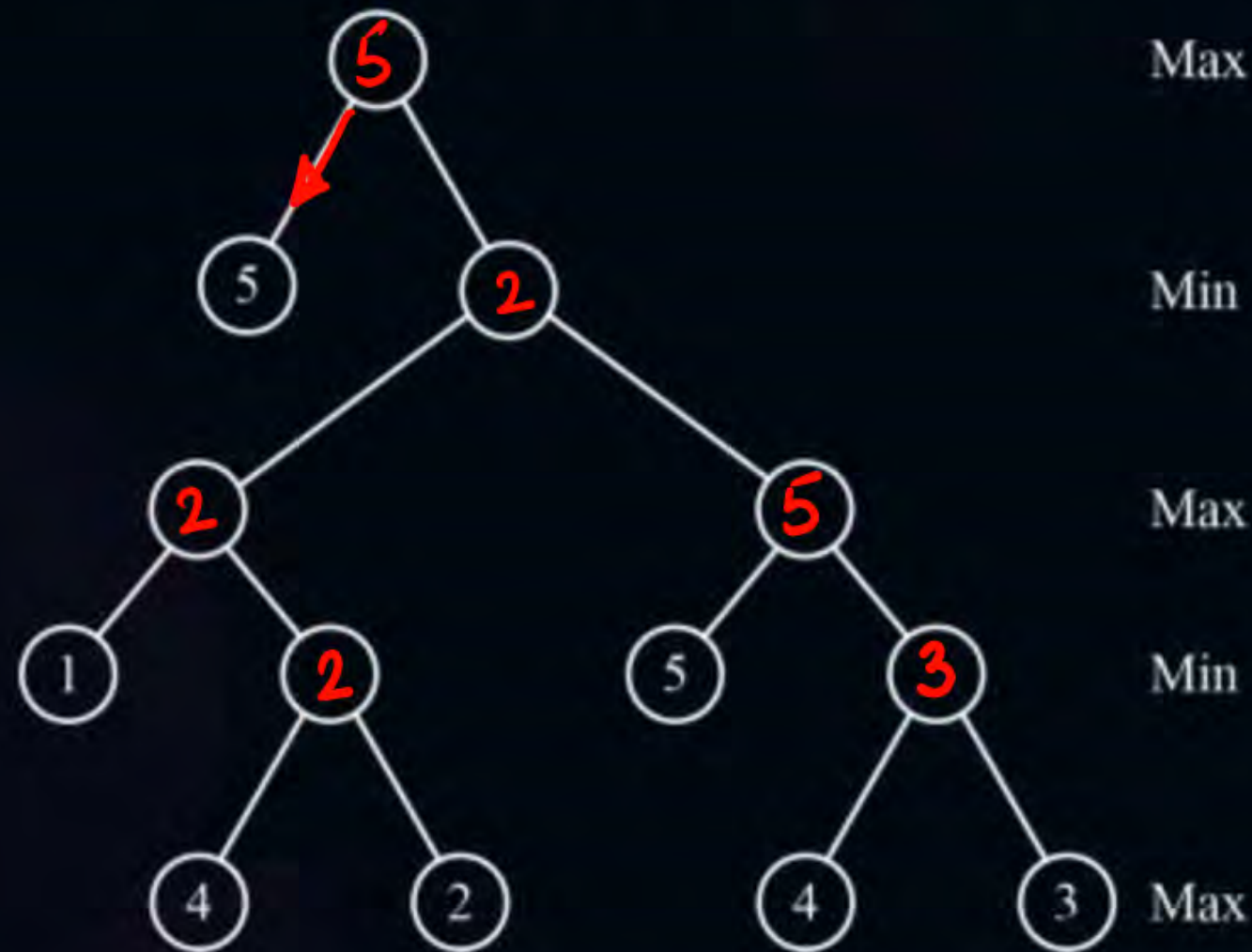
Adversarial search

DPP 02 Discussion Notes



By - Siddharth Sabharwal Sir

#Q. What is the minimax value of the root node for the game tree below?



Assuming that it performs a depth-first search that always generates the leftmost child node first and a loss (and win) of MAX (and MIN) corresponds to a value of $-\infty$ (and ∞ , respectively).

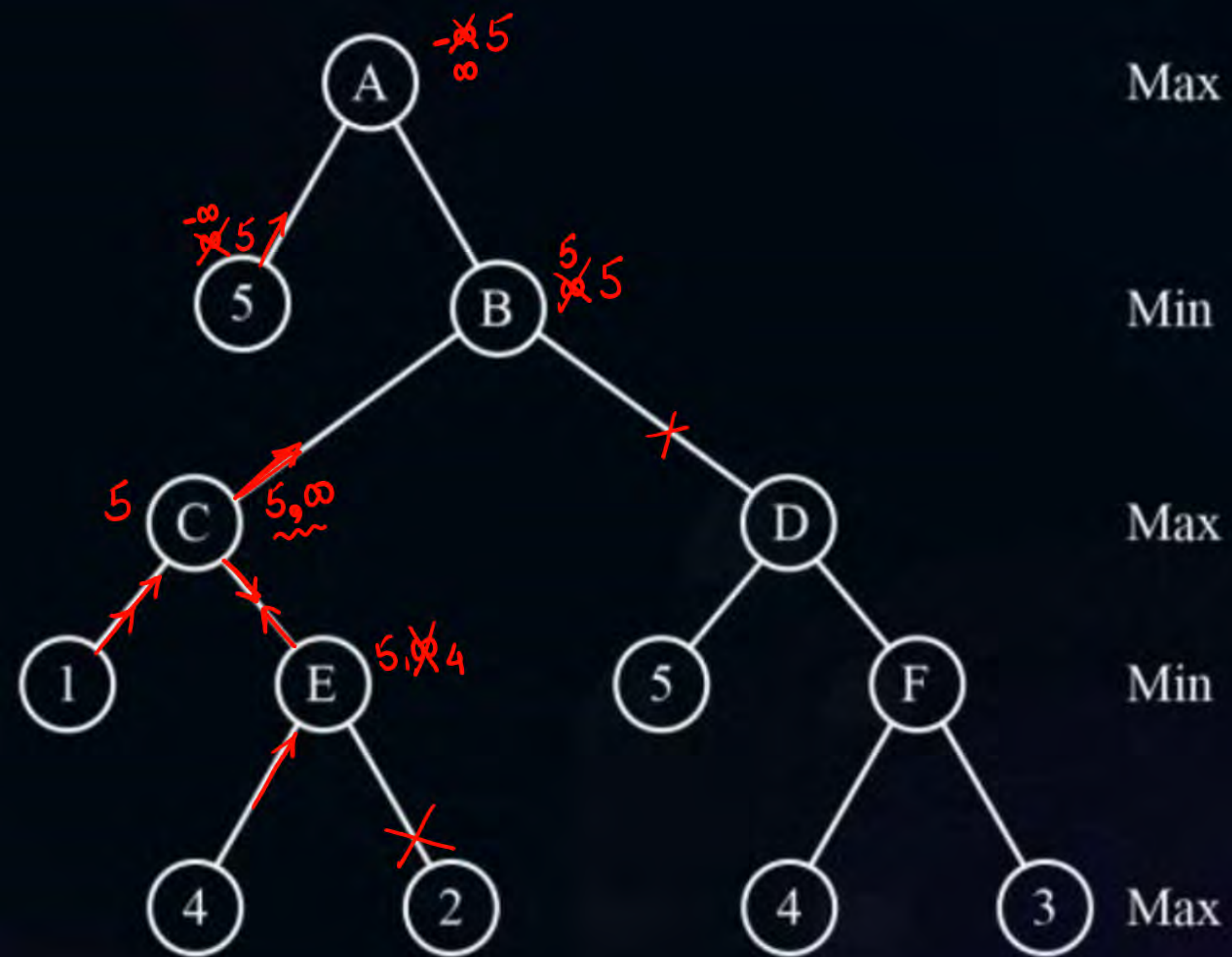
[NAT]

#Q. Consider the below search tree

If we apply α - β pruning method to above tree, then which of the node(s) will get pruned?

Assuming that it performs a depth-first search that always generates the leftmost child node first and a loss (and win) of MAX (and MIN) corresponds to a value of $-\infty$ (and ∞ , respectively).





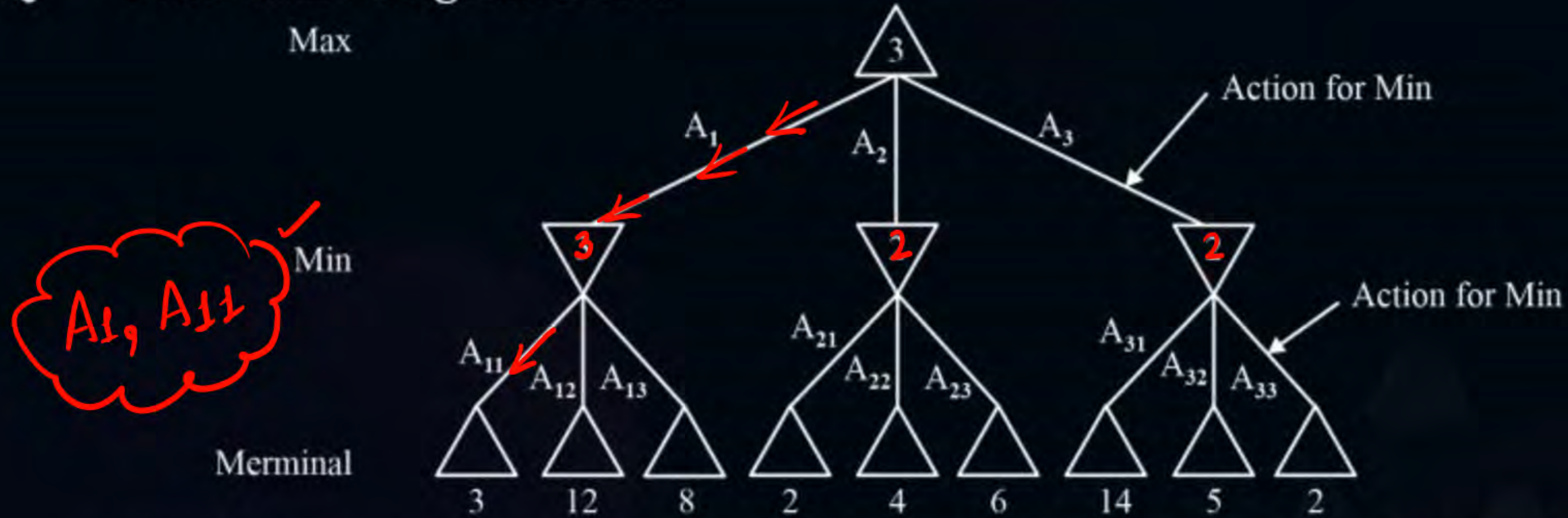
Sol 5 //

[MCQ]

#Q. What is the time complexity of the Minimax algorithm in terms of the branching factor (b) and the depth of the search tree (d)?

- ☒ A $O(b^d)$ ✓
- ☐ B $O(b.d)$
- ☐ C $O(b^{d/2})$
- ☐ D $O(b^b)$

#Q. Draw same diagram 5 also



Consider the following game tree where MAX and MIN take turns. The terminal node values are given at the bottom of the tree.

Assume both players play optimally.

- What is the value at the root node? 3
- Which action should MAX take at the root node?

A Value = 3, Action = A_1 ✓

B Value = 2, Action = A_2

C Value = 2, Action = A_3

D Value = 4, Action = A_2

[MSQ]

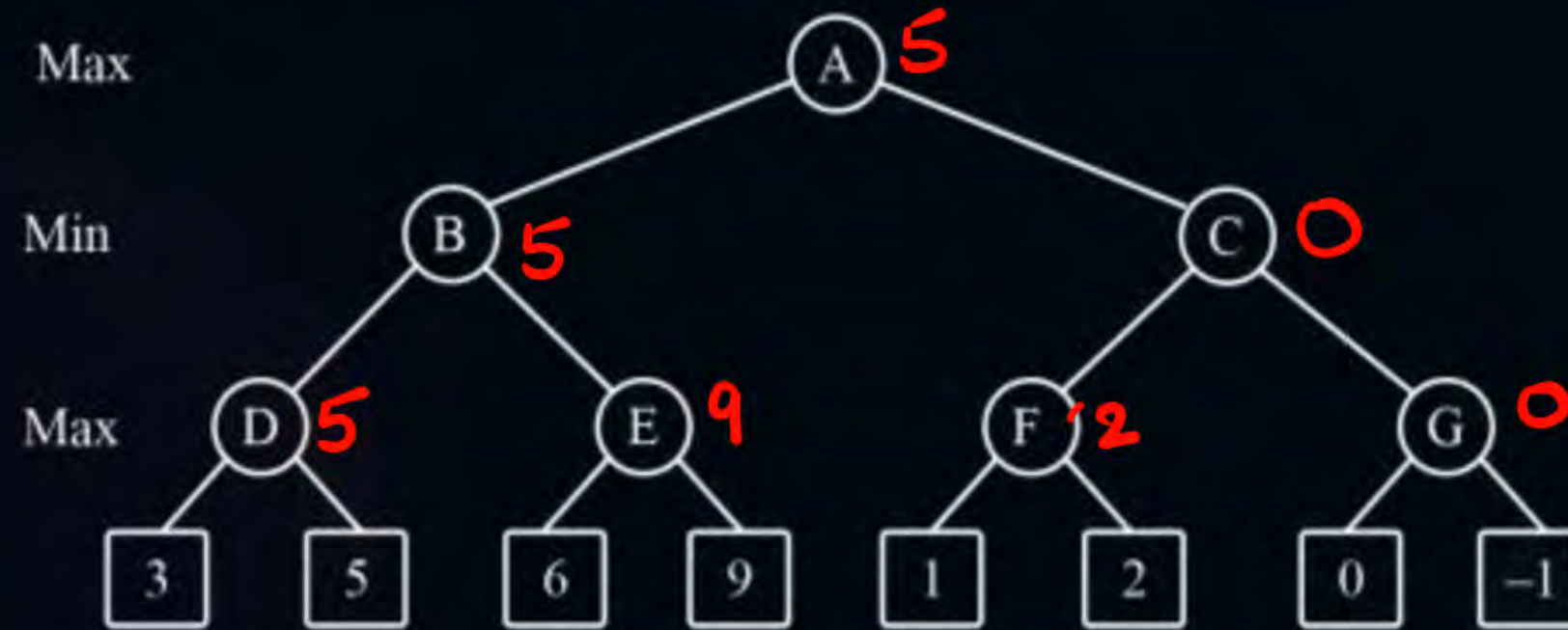
#Q. Which of the following is/are false regarding the alpha-beta pruning for the mini-max search algorithm?

done in class

- ☐ A It can potentially lead to suboptimal solutions compared to mini-max search without any pruning *✗*
- ☒ B *✗* It is guaranteed to improve the running time in-comparison to the mini-max search without any pruning
- ☒ C *✓* The order in which nodes are visited affects the amount of pruning
- ☐ D If the successors of a node are chosen randomly, the time complexity (on average) is $O(b^{3m/4})$ *$(b^{3m/4})$*

[MCQ]

#Q. In the following two-ply game tree, the terminal nodes show the utility values computed by the utility function. Use the Minimax algorithm to compute the utility values for other nodes in the given game tree.



What will be the utility value for Node A, B and C?

A

$A = 5, B = 5 \text{ \& } C = 0$



B

$A = 0, B = 5 \text{ \& } C = 0$

C

$A = 0, B = 9 \text{ \& } C = 2$

D

$A = -1, B = 3 \text{ \& } C = -1$

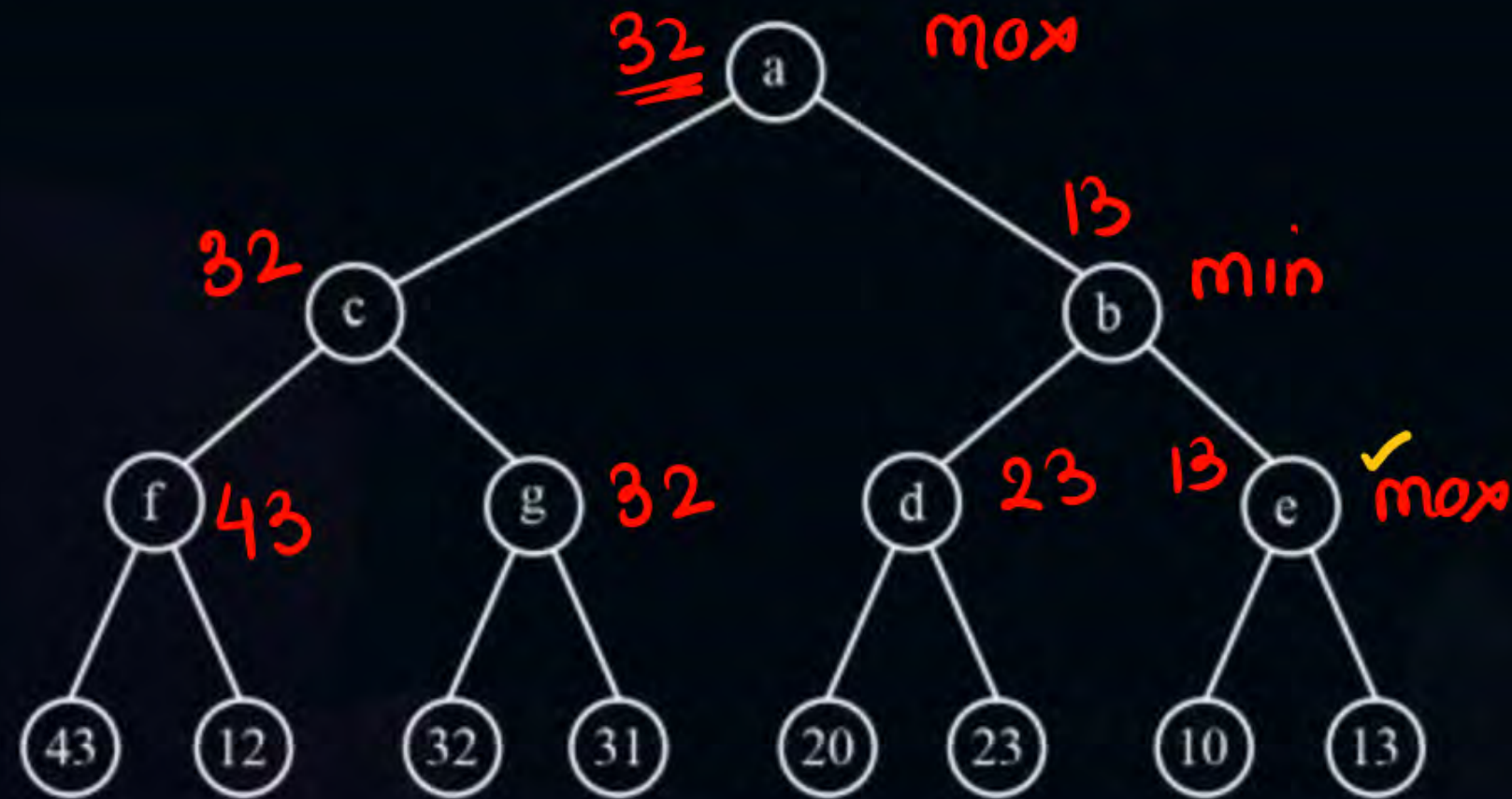
[MCQ]

#Q. You know from experience that the value of a particular state in a MiniMax search tree lies between x and y ($x < y$). How can you use this information to optimize your alpha-beta search?

done in class

- A** Set $\alpha = x$ and $\beta = y$
- B** Set $\alpha = y$ and $\beta = x$
- C** Return value of state as $(x + y)/2$
- D** This information cannot be used to optimize the procedure

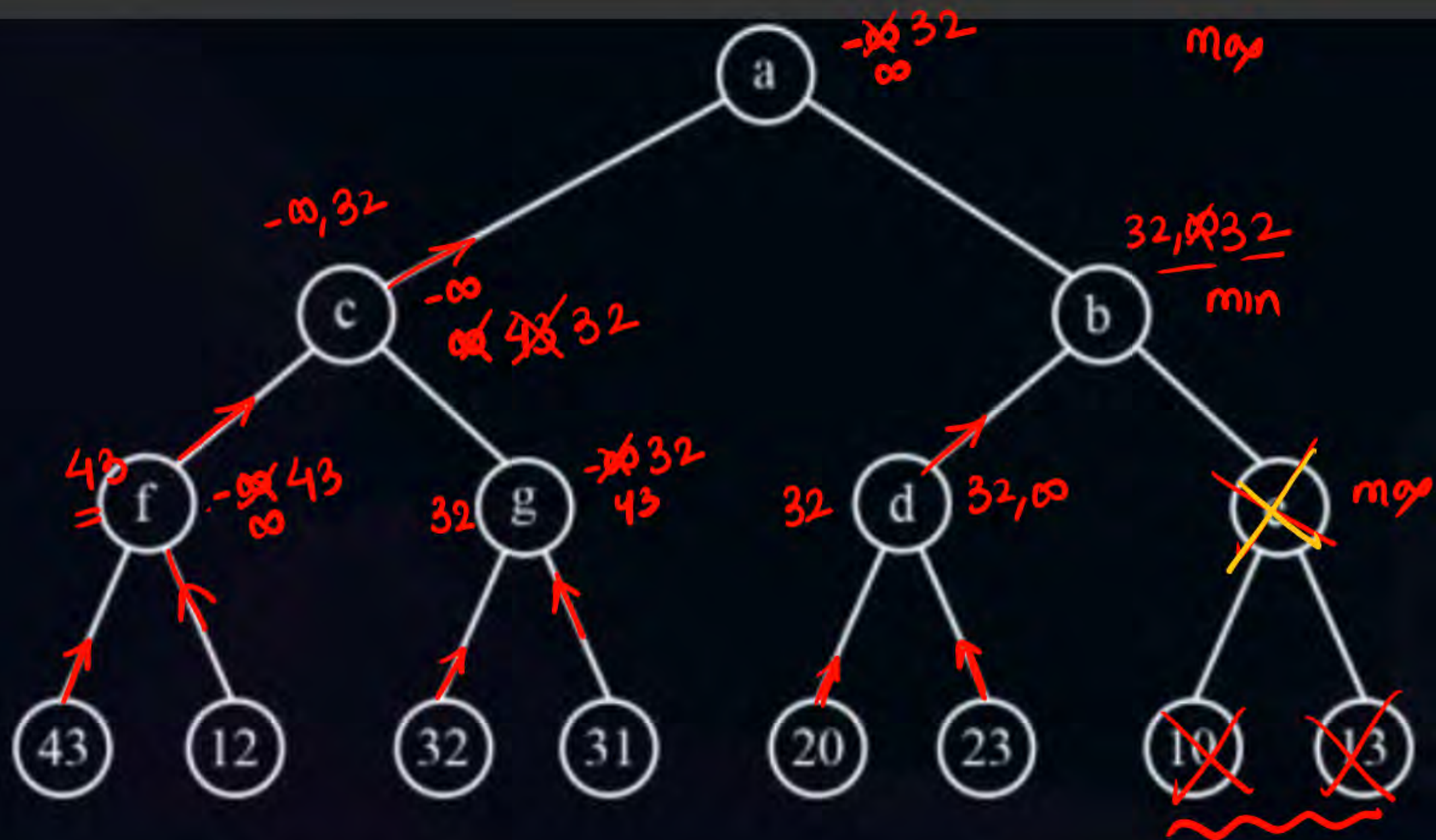
- #Q. What is the value of the root node after MiniMax search over the given tree?
The root node (labelled 'a') is a max node.



#Q. How many nodes (including leaves and internal nodes) will an alpha-beta pruning based MiniMax search visit in the given tree, assuming each successor is ordered left to right?

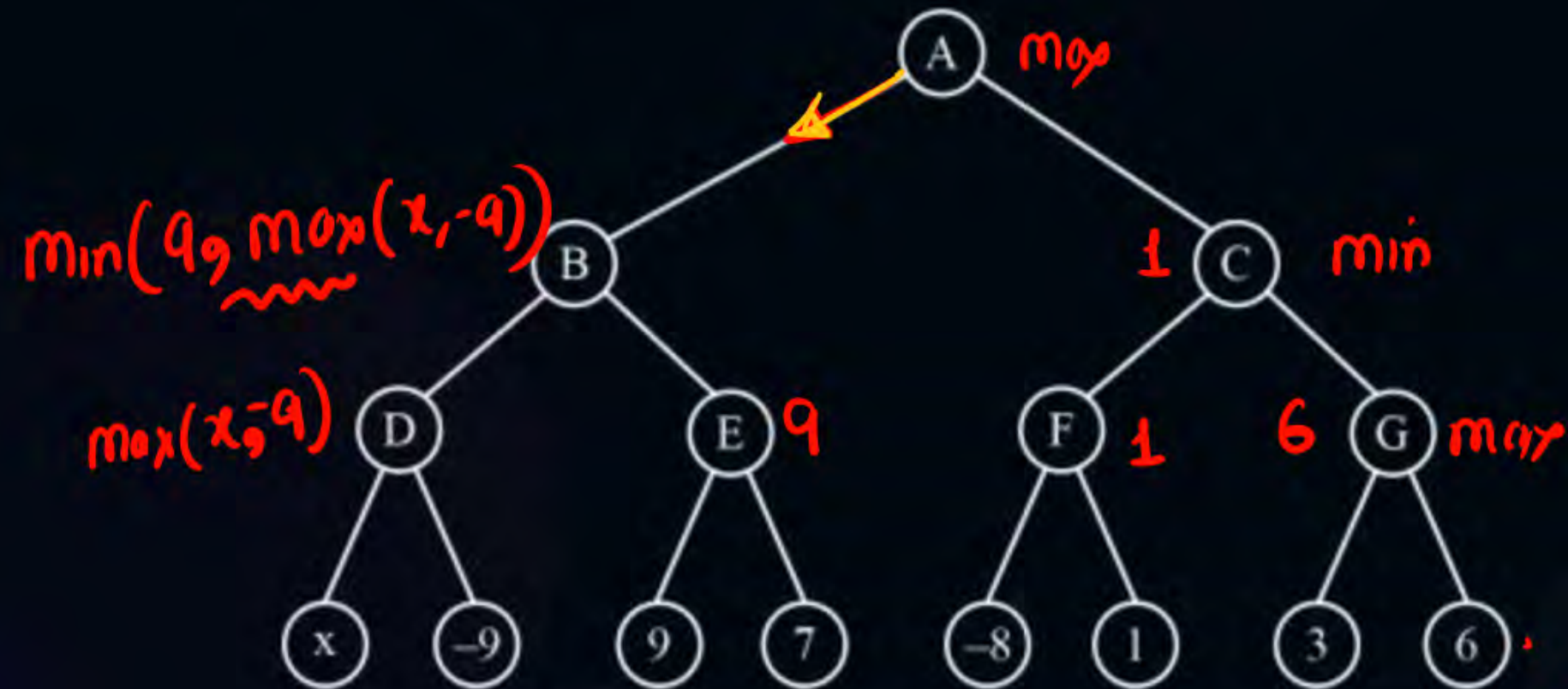
The root node (labelled 'a') is a max node.





[MCQ]

#Q. Player 1 (max node) chooses the left action for which of the following value(s) of x . Assume optimal play and Player 1's turn at Node A



A 2 ✓

C 4 ✓

B -7

D -1

$$\min(9, \max(x, -9)) > 1$$

$x > -9$ $\min(9, x) > 1$

$1 < x < 9$, $x > 1$

$$\min(q, \max(x, -q)) > 1$$

$$\text{if } x > -q, \quad \min(q, x) > 1$$

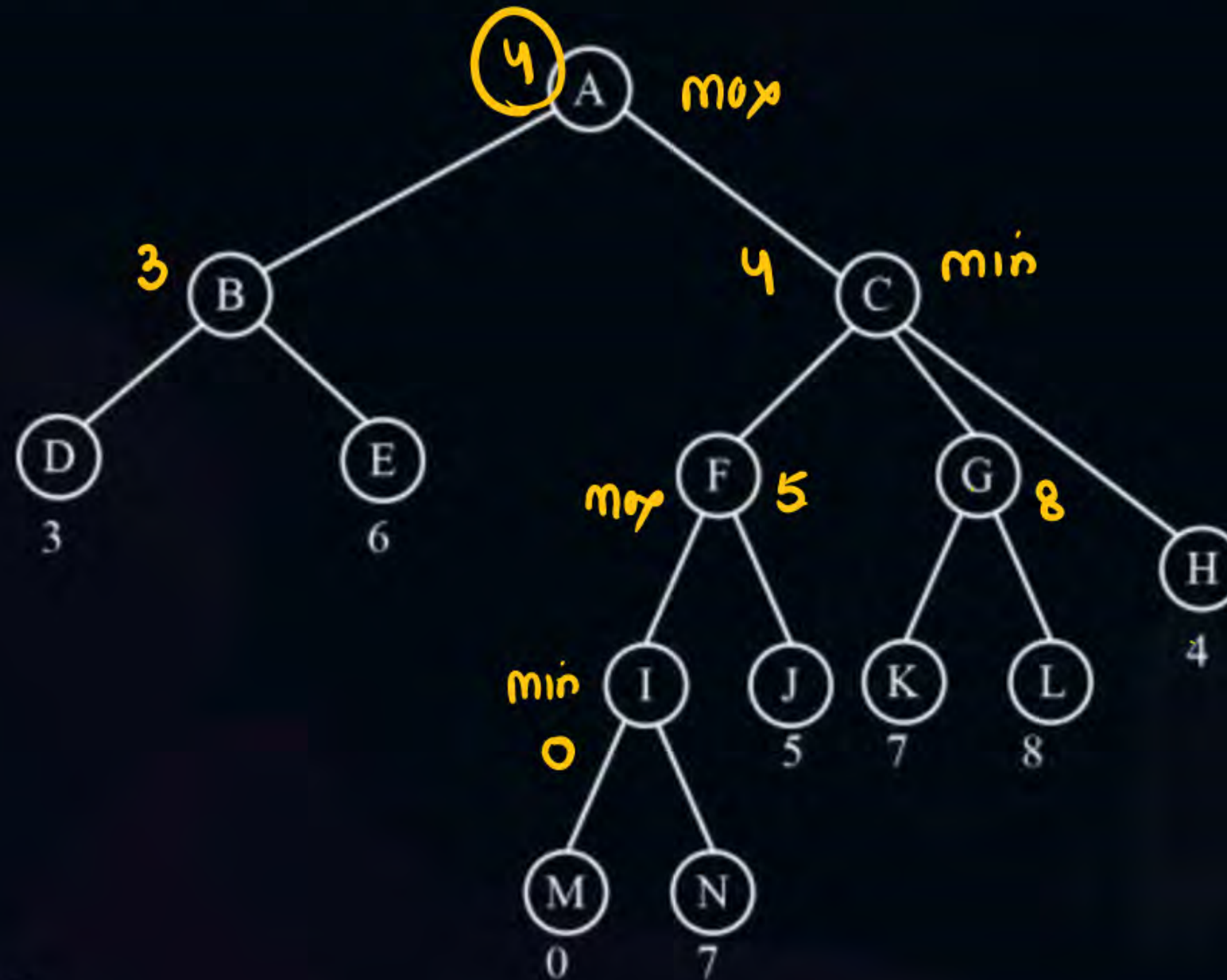
$$1 < x < q, \quad \text{true}$$

$$x > q, \quad q > 1 + \text{true}$$

$$\boxed{x > 1} \checkmark$$

[NAT]

#Q. Perform minmax on the following tree-



Explain each step.

Perform Minimax Search on the given game tree.

[MCQ]

#Q. In adversarial search, $\alpha - \beta$ pruning can be applied to game trees of any depth where α is the (m) value choice we have formed so far at any choice point along the path for the MAX player and β is the (n) value choice we have formed so far at any choice point along the path for the MIN player.

Which ONE of the following choices of (m) and (n) and makes the above statement valid?

** done in class **

A

(m) highest, (n) = highest

B

(m) = lowest, (n) = highest

C

(m) = highest, (n) = lowest

D

(m) = lowest, (n) = lowest

#Q. Which of the following statements are true?

- i. Minimax search is breadth-first^x; it processes all the nodes at a level before moving to a node in next level.
- ~~ii.~~ The effectiveness of the alpha-beta pruning is highly dependent on the order in which the states are examined ✓
- ~~iii.~~ The alpha-beta search algorithms computes the same optimal moves as minimax algorithm
- iv. Optimal play in games of imperfect information does not require reasoning about the current and future belief states of each player

free

Choose the correct answer from the options given below:

A ~~(i)~~ and (iii) only

B ~~(i)~~ and (iv) only

C (ii) and (iii) only

D (iii) and (iv) only

[MCQ]

#Q. In the context of adversarial search, $(\alpha-\beta)$ pruning is an optimization technique applied to the Minimax algorithm to reduce the number of nodes that need to be evaluated.

Which of the following statements about Alpha-Beta pruning is/are true?

- ☐ A $\alpha-\beta$ pruning is generally faster than minimax, but loses the guarantee of optimality. ✗
- ☒ B $\alpha-\beta$ pruning can reduce computation time by pruning portions of the game tree. ✓
- ☒ C $\alpha-\beta$ pruning always returns the same value as minimax for all nodes on the leftmost edge of the tree, assuming successor game states are expanded from left to right. ✓
- ☐ D $\alpha-\beta$ pruning always returns the same value as minimax for all nodes of the tree. ✗

[illegible]

- Perform the Minimax algorithm to compute the value at the root node.
- Identify which branch the root node will choose. ✓ ✓

(Hint: At MIN nodes, choose the minimum value; at MAX nodes, choose the maximum value.)

A

Root value = 4, Branch = Right Subtree

B

Root value = 1, Branch = Left Subtree

C

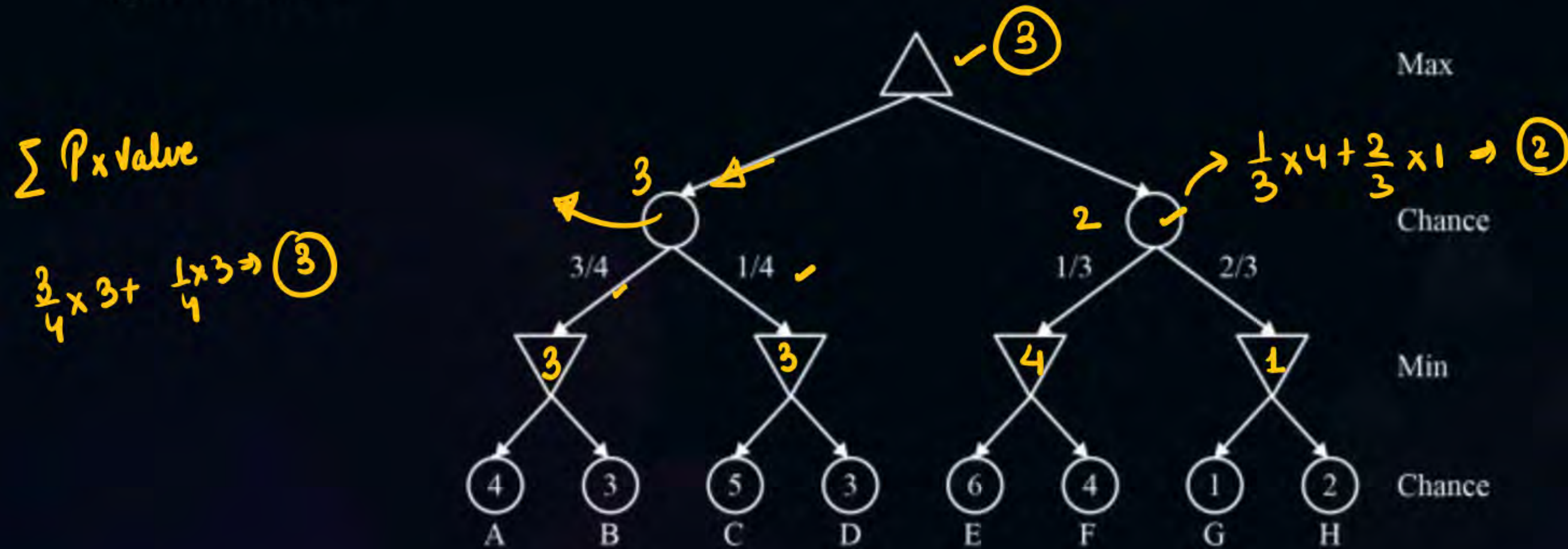
Root value = 2, Branch = Middle Subtree ✓✓

D

Root value = 5, Branch = Left Subtree

[NAT]

#Q. Consider the following two-player game tree in which the game has an element of chance, which is shown by the so-called probability nodes in the game tree.



What is the correct value for the top MAX node if one applies the expect minimax algorithm? (3)

[MCQ]

#Q. Let h_1 and h_2 be two admissible heuristics used in A^* search.
Which ONE of the following expressions is always an admissible heuristic?

- A $h_1 + h_2$ ✗
- B $h_1 \times h_2$ ✗
- C $h_1/h_2, (h_2 \neq 0)$ ✗
- D $|h_1 - h_2|$ ✓

[MCQ]

#Q. The state graph shows the action cost along the edges and the heuristic function h associated with each state.

$S \rightarrow A \rightarrow B$



Suppose A^* algorithm is applied on this state graph using priority queue to store the frontier. In what sequence are the nodes expanded?

A

S, A, E, C, B, D, G

✗

B

S, E, A, C, B, D, G

✗

D

S, A, B, E, C, D, G

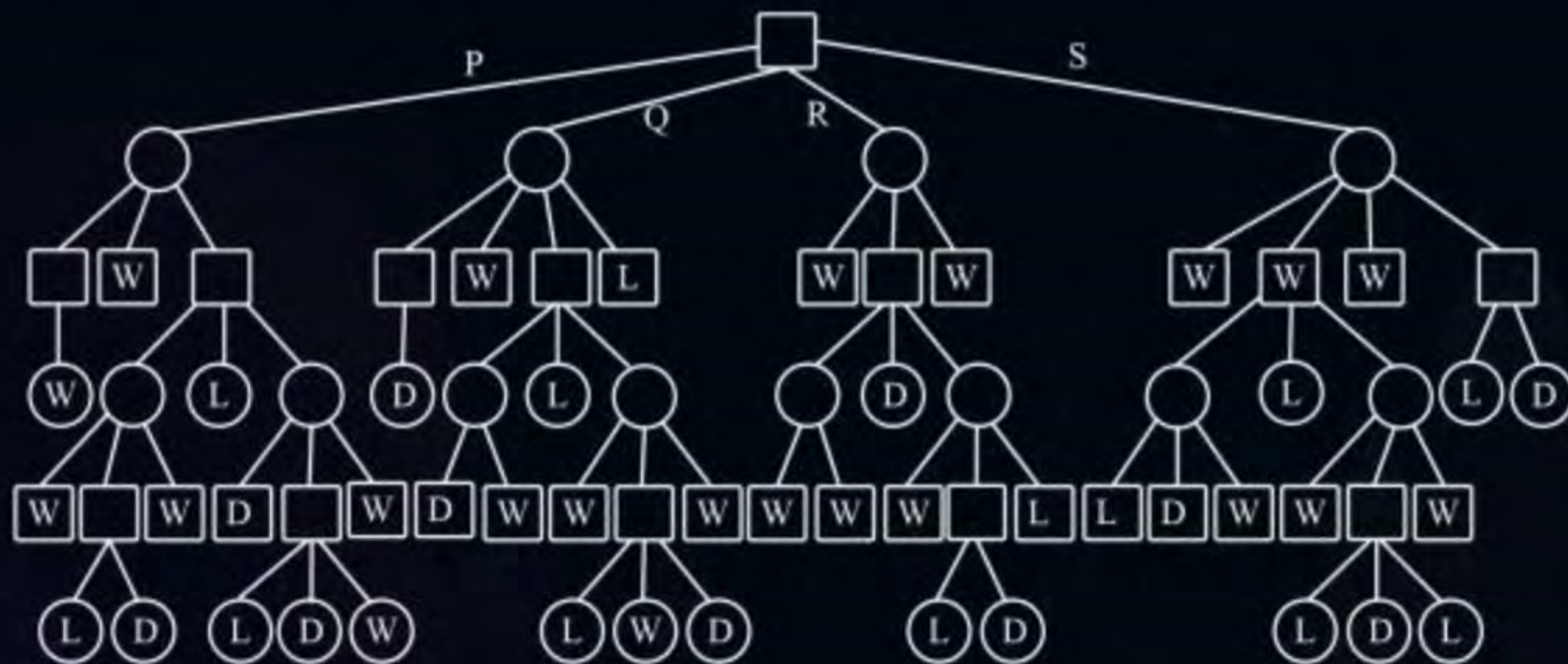
C

S, A, E, B, C, D, G

S
 $A \ 6$
 $E \ 7$
 $B \ 5$

#Q. The figure shows a game tree with evaluations W (win), L (loss) and D (draw) from Max's perspective. In this game tree the labels P, Q, R, S indicate strategies/moves.

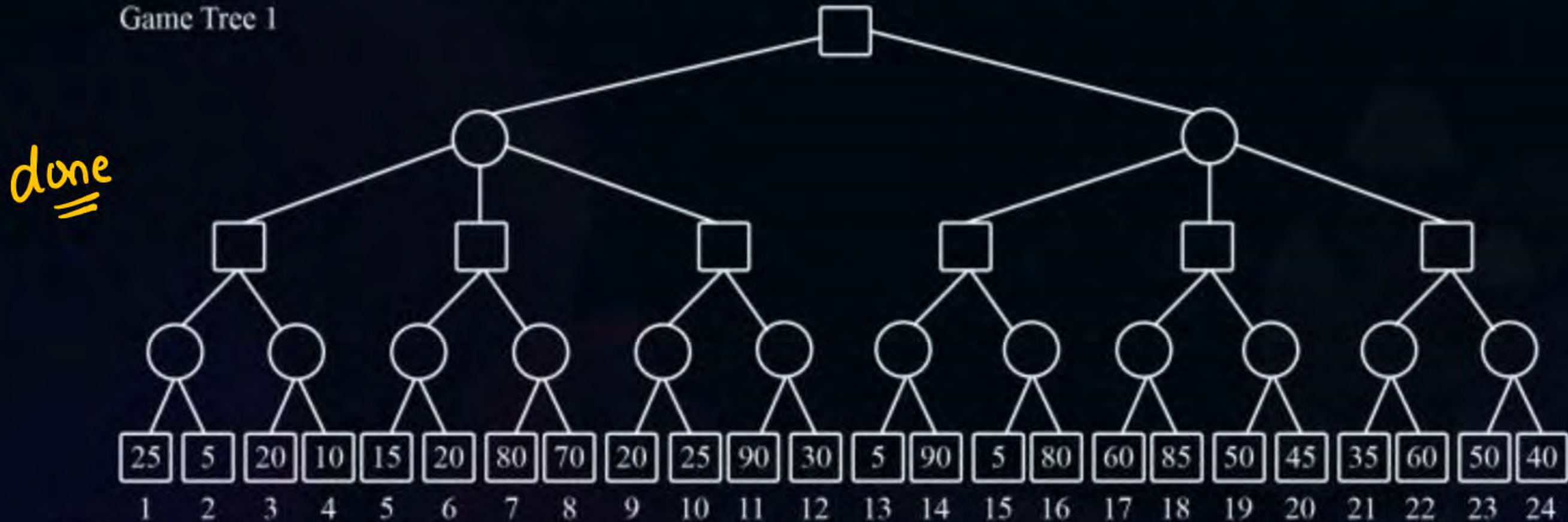
done



What is the outcome (W,L or D) of the game when both players play perfectly?

- #Q. The figure shows a 4-ply game tree with evaluation function values at the horizon. The nodes in the horizon carry reference numbers (sequence numbers going left to right) at the bottom. Use these reference numbers when you want to enter a list of nodes. Where necessary assume top-down and left-to-right node order. Use this game tree to answer the following questions.

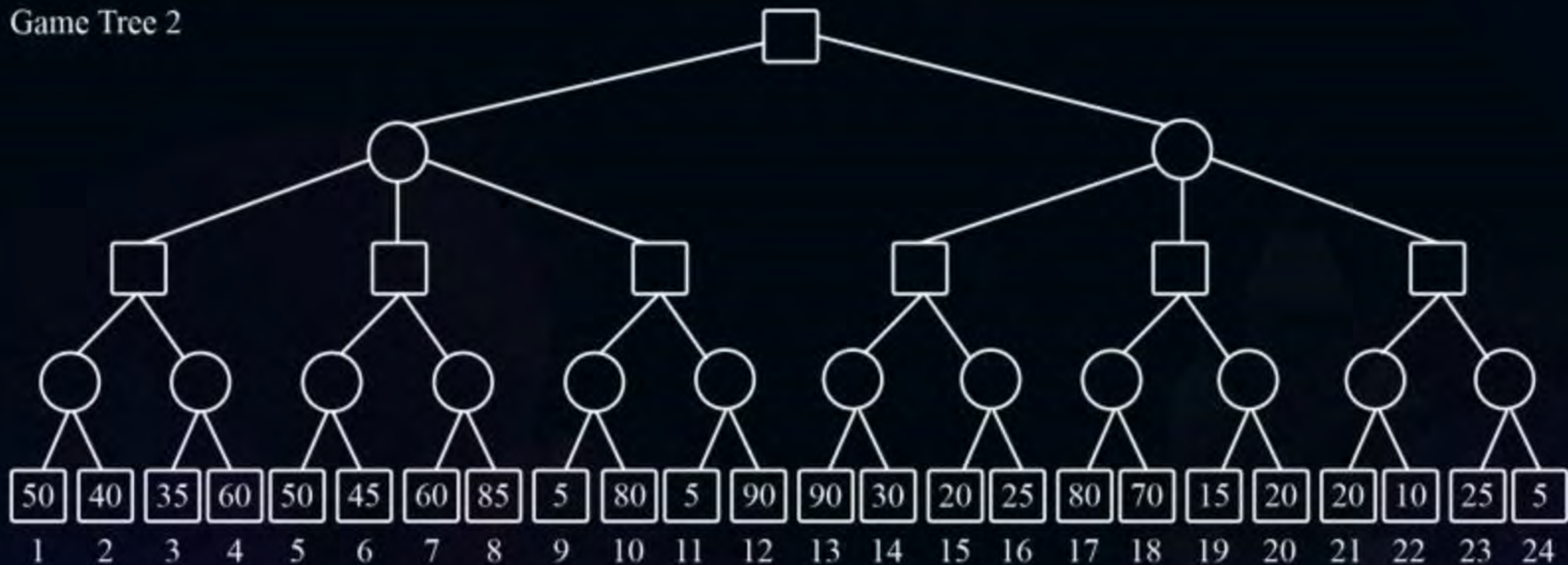
Game Tree 1



What is the MinMax value of the game?

#Q. What can you say about Game Tree 1 and Game Tree 2? Select all the correct options.

Game Tree 2



- A Each tree represents a different game
- B The possible games played in the two trees are the same
- C The MinMax values are the same
- D The MinMax values are different
- E The amount of pruning done by AlphaBeta algorithm is the same in both game trees

[MCQ]

#Q. For more complex games, such as chess or checker the AND/OR search to termination is out of question. Which of the following statements are true?

1. Our goal in searching such a game tree might be, instead, merely to find a good first move. ✓
2. Extract from the search graph an estimate of the 'best' first move

- ☒ A Both 1 and 2
- ☐ B Only 2
- ☐ C Only 1
- ☐ D Both 1 and 2 are false.

Tricky

Start → end

we just want to find best move based on current situation of game

[MCQ]

#Q. Using alpha-beta pruning, in the best case, you examine only $(2b)^{\lceil d/2 \rceil}$ leaf nodes. The result is you can search _____ as deep as minimax!

Minimax b^d
 α - β pruning $b^{d/2}$

- ☒ A Equal
- ☐ B Twice ✓
- ☐ C Half
- ☐ D One-fourth



THANK - YOU